

THE GIANT STEPS

- Building vastly more powerful launch vehicles will give India more options in space exploration
- Launch cost is critical for frequent space missions. Having reusable launch vehicles cuts costs by over 90 per cent and increases productivity
- India should aim to land astronauts on the Moon by 2050. It is a surefire way to announce its advent as a top space power
- For India to become a space superpower, major technology development should be outsourced to the private sector

Back to the Moon with humans: The US is all set to go back to the Moon with humans, followed by China. The Artemis Program will put American astronauts on the Moon in 2026. The US has been there before, but this time it will not be to merely visit but to set up a base, which will be staffed by astronauts 365 days a year. It may be supported by multiple governments, like the International Space Station, or it could be private—operated by Blue Origin, owned by Jeff Bezos.

The search for life on Mars: We have learnt a lot about Mars from multiple rover and orbiter missions since 1996. We have learnt about the presence of water at the poles and in the Martian subsurface, we have learnt about Martian earthquakes and dust storms. It is possible that in future, there may be evidence of past life—microscopic in nature—on Mars. The search for life on Mars is expected to continue for the next couple of decades and will remain a frontier of space exploration.

Oceans in the outer solar system: Many of Jupiter and Saturn's moons, like Europa and Enceladus, have oceans below the crust. One of Jupiter's moons, Europa, has an ocean about 100 km below its surface. It might contain two-three times more water than all of the Earth's oceans combined. Since life on Earth originated in the oceans, is it possible that Europa and Enceladus's oceans could have microscopic life? Excitement in the outer Solar System has begun: NASA launches the Europa Clipper later this year. NASA's Dragonfly Mission to Titan (a moon of Saturn) is scheduled to launch in 2028.

Given this kaleidoscope of developments in space exploration, how can India strategically and operationally become a key player and a space superpower? These are some areas to focus on.

More powerful launch vehicles: India needs to develop more powerful launch vehicles. Its most powerful launch vehicle, the GSLV, can deliver 6,000 kg to low earth orbit (LEO). In contrast, the NASA's Saturn 5 (from the Apollo Program) and Starship (from SpaceX) can deliver 140,000 kg and 250,000 kg respectively to a LEO. For landing humans on the Moon, China is developing its Long March 10 that will deliver about 70,000 kg to LEO. One way for India to become a superpower would be to land an astronaut on the Moon in the next couple of decades. Having a more powerful launch vehicle would enable this. A powerful launch vehicle is a key capability that will provide India with multiple options. For example, it will enable India to send missions to the Outer Solar System: to study Moons like Europa, Enceladus or Titan. It will make possible missions to planets like Jupiter, Saturn and Uranus.

Reusable launch vehicles: The launch vehicles should be reusable. To frequently undertake space missions, the cost of launch is critical. For a manned spaceflight, one mission can cost over $1 billion. If the launch vehicle is reused, the cost of launch can be decreased by over 90 per cent. Be it a manned or robotic spaceflight, missions to other planets or Earth satellites, the cost of launch is approximately half the total cost. If launch costs can be reduced, cost of savings and increase in productivity are huge. The Falcon launch vehicles developed by SpaceX have been a game-changer because they are reusable.

Human spaceflight: After sending robotic spacecrafts to Moon and Mars, a manned spaceflight is the next logical step for India. Manned spaceflight is harder and more expensive. India should aim to land astronauts on the Moon by 2050.

Implementation model: Despite being a trailblazer in spacecraft missions in the past 20 years, ISRO faces substantial odds to attract and retain talent. The private sector, including MNCs like Google and Apple, pay several multiples more than the salary at ISRO. There are questions of bureaucratic procedures and timelines. These issues can be addressed more effectively by the private sector. Can major technology development projects be outsourced to the private sector for more effective implementation?

When its space shuttles were retired, NASA experimented with a private-public partnership programme called the Commercial Crew Program. NASA gave SpaceX and Boeing about $2.6 billion and $4.2 billion to develop a spacecraft which would replace the shuttles. If they were successful, NASA promised to pay them a fee to ferry astronauts to the International Space Station and back. This creates an assured market and a viable business model for SpaceX and Boeing. A decade later, this strategy seems to have worked for NASA: SpaceX has delivered the Crew Dragon spacecraft, a robust alternative to the shuttles, in around a decade. Boeing's Starliner spacecraft is going through the last developmental challenges. ISRO has been increasingly partnering with the private sector; major technology development like the next generation launch vehicle could be outsourced to the Indian private sector. If this works, India can leapfrog its way to become a space superpower by delivering on difficult technology tasks. ■

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